

8.3.5 Practical Conjugate Gradient Method algorithm

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fit width



We have noted that $p^{(k)} = r^{(k)} + \gamma_k p^{(k-1)}$. Since $p^{(k)}$ is A-conjugate to $p^{(k-1)}$ we find that

$$p^{(k-1)T} A p^{(k)} = p^{(k-1)T} A r^{(k)} + \gamma_k p^{(k-1)T} A p^{(k-1)}$$

so that

$$\gamma_k = -p^{(k-1)T} A r^{(k)} / p^{(k-1)T} A p^{(k-1)}.$$

This yields the first practical instantiation of the Conjugate Gradient method, given in [Figure 8.3.5.1](#).

```
Given : A, b
x(0) := 0
r(0) := b
k := 0
while r(k) ≠ 0
  if k = 0
    p(k) = r(0)
  else
    γk := -p(k-1)T A r(k) / p(k-1)T A p(k-1)
    p(k) := r(k) + γk p(k-1)
  endif
  αk := p(k)T r(k) / p(k)T A p(k)
  x(k+1) := x(k) + αk p(k)
  r(k+1) := r(k) - αk A p(k)
  k := k + 1
endwhile
```

Figure 8.3.5.1. Conjugate Gradient Method.

Homework 8.3.5.1. In [Figure 8.3.5.1](#) we compute

$$\alpha_k := \frac{p^{(k)T} r^{(k)}}{p^{(k)T} A p^{(k)}}.$$

Show that an alternative formula for α_k is given by

$$\alpha_k := \frac{r^{(k)T} r^{(k)}}{p^{(k)T} A p^{(k)}}.$$

[Hint](#) [Solution](#)

We need to show that $p^{(k)T} r^{(k)} = r^{(k)T} r^{(k)}$.

$$\begin{aligned} p^{(k)T} r^{(k)} &= \langle r^{(k)} + \gamma_k p^{(k-1)}, r^{(k)} \rangle \\ &= \langle r^{(k)} + \gamma_k p^{(k-1)} \rangle^T r^{(k)} \\ &= \langle \text{distribute} \rangle \\ &= r^{(k)T} r^{(k)} + \gamma_k p^{(k-1)T} r^{(k)} \\ &= \langle p^{(k-1)T} r^{(k)} = 0 \rangle \\ &= r^{(k)T} r^{(k)}. \end{aligned}$$

Use the fact that $p^{(k)} = r^{(k)} + \gamma_k p^{(k-1)}$ and the fact that $r^{(k)}$ is orthogonal to all previous search directions to show that $p^{(k)T} r^{(k)} = r^{(k)T} r^{(k)}$.

The last homework justifies the refined Conjugate Gradient Method in [Figure 8.3.5.2](#) (Left).

Given : A, b	Given : A, b
$x^{(0)} := 0$	$x^{(0)} := 0$
$r^{(0)} := b$	$r^{(0)} := b$
$k := 0$	$k := 0$
while $r^{(k)} \neq 0$	while $r^{(k)} \neq 0$
if $k = 0$	if $k = 0$
$p^{(k)} = r^{(0)}$	$p^{(k)} = r^{(0)}$
else	else
$\gamma_k := -(p^{(k-1)T} A r^{(k)}) / (p^{(k-1)T} A p^{(k-1)})$	$\gamma_k := (r^{(k)T} r^{(k)}) / (r^{(k-1)T} r^{(k-1)})$
$p^{(k)} := r^{(k)} + \gamma_k p^{(k-1)}$	$p^{(k)} := r^{(k)} + \gamma_k p^{(k-1)}$
endif	endif
$\alpha_k := \frac{r^{(k)T} r^{(k)}}{p^{(k)T} A p^{(k)}}$	$\alpha_k := \frac{r^{(k)T} r^{(k)}}{p^{(k)T} A p^{(k)}}$
$x^{(k+1)} := x^{(k)} + \alpha_k p^{(k)}$	$x^{(k+1)} := x^{(k)} + \alpha_k p^{(k)}$
$r^{(k+1)} := r^{(k)} - \alpha_k A p^{(k)}$	$r^{(k+1)} := r^{(k)} - \alpha_k A p^{(k)}$
$k := k + 1$	$k := k + 1$
endwhile	endwhile

Figure 8.3.5.2. Alternative Conjugate Gradient Method algorithms.

Homework 8.3.5.2. For the Conjugate Gradient Method discussed so far,

- Show that

$$r^{(k)T} r^{(k)} = -\alpha_{k-1} r^{(k)T} A p^{k-1}.$$

- Show that

$$p^{(k-1)T} A p^{(k-1)} = r^{(k-1)T} r^{(k-1)} / \alpha_{k-1}.$$

▼ [Hint](#) ▼ [Solution](#)

$$\begin{aligned}
 r^{(k)T} r^{(k)} &= r^{(k)T} r^{(k-1)} - \alpha_{k-1} r^{(k)T} A p^{k-1} = -\alpha_{k-1} r^{(k)T} A p^{k-1}. \\
 &= p^{(k-1)T} A p^{(k-1)} \\
 &= (r^{(k-1)} - \gamma_{k-1} p^{(k-2)})^T A p^{(k-1)} \\
 &= r^{(k-1)T} A p^{(k-1)} \\
 &= r^{(k-1)T} (r^{(k-1)} - r^{(k)}) / \alpha_{k-1} \\
 &= r^{(k-1)T} r^{(k-1)} / \alpha_{k-1}.
 \end{aligned}$$

Recall that

$$r^{(k)} = r^{(k-1)} - \alpha_{k-1} A p^{(k-1)}. \quad (8.3.4)$$

and rewrite [\(8.3.4\)](#) as

$$A p^{(k-1)} = (r^{(k-1)} - r^{(k)}) / \alpha_{k-1}.$$

and recall that in the previous iteration

$$p^{(k-1)} = r^{(k-1)} - \gamma_{k-1} p^{(k-2)}.$$

From the last homework we conclude that

$$\gamma_k = -(p^{(k-1)T} A r^{(k)}) / (p^{(k-1)T} A p^{(k-1)}) = r^{(k)T} r^{(k)} / r^{(k-1)T} r^{(k-1)}.$$

This is summarized in on the right in [Figure 8.3.5.2](#).